

Advanced Applied Econometrics: Structural econometric techniques in labor economics

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Part II: Structural Micro - IO and Labor

Lectures:

- June 5th: Family labor supply: static discrete choice (v. Soest, 1995)
- **IO**
 - ▶ June 12th & 19th: Static demand estimation with market-level data (Berry (1994), BLP (1995))
 - ▶ June 26th: Dynamic discrete choice models: Rust (1987) bus engine replacement model
- **Labor**
 - ▶ July 3rd: Dynamic occupational choice models (Keane and Wolpin, 1997)
 - ▶ June 10th: Dynamic labor supply and welfare reform (Blundell et al., 2016)
- **Exam:** July 17th, 9am; Elinor Ostrom Hall; duration 2hrs

Part II: IO - Exercises

Exercises:

- Exercise 1 : Static demand estimation with market-level data I
 - ▶ Introduction of Berry/BLP framework: estimation of demand for differentiated products
 - ▶ Logit CCP, dataset simulation, IV/GMM estimator setup, contraction mapping
- Exercise 2: Static demand estimation with market-level data II
 - ▶ Estimation of Static Differentiated Product Demand Models
 - ▶ Application of techniques introduced in Exercise 1
- Exercise 3: Dynamic Discrete Choice Estimation - Rust (1987)
 - ▶ Harry Zurcher's bus engine replacement model, intro to dynamic programming
 - ▶ Implementation of Rust's nested fixed point algorithm for estimation of dynamic discrete choice models
- **Graded PS 3:** More complex version of Rust (1987)

Part II: Labor Economics - Exercises

Exercises:

- Exercise 0 (today): Python demo
- Exercise 1: Static discrete choice of labor supply
 - ▶ Static choice of working hours from finite set of alternatives
 - ▶ Simulation of simple dataset; recovering parameters using maximum likelihood estimation
- Exercise 2: Dynamic discrete choice of labor supply I
 - ▶ Simple finite-horizon Robinson economy with deterministic evolution of states
 - ▶ Implementation of state-space, indexer and model solution with discrete choice dynamic programming techniques (backwards induction)
- **Graded PS 4:** Dynamic discrete choice of labor supply II
 - ▶ More complex state-space; introduction of uncertainty/stochastic elements
 - ▶ Expectation maximization, multidimensional numerical integration, estimation using simulated method of moments

Organization

Graded Problem Set 4 - Labor:

- Due date: **August 15**
- Additional tutorial: coordinate date!

Exam:

- Date: **July 17, 9am**; Elinor Ostrom Hall; duration 2hrs
- All material covered in the course is relevant for the exam.
- 1/2 Part I (FW), 1/2 Part II: IO (HU) & labor (PH)
- On Part II: There will be no explicit questions on Python programming, however, you need to be able discuss the conceptual implementation of the models and methods discussed in the problem sets.
- Materials allowed:
 - ▶ Hand-written or printed notes: 10 pages, double-sided, A4. ****No**** lecture slides/papers etc.
 - ▶ **No** electronic devices
 - ▶ Non-programmable calculator
- ****Master students**** receive the same exam questions. Admission requires passing grade on 3/4 problem sets.

In-class presentations

- Every week, 3 teams will present their solutions to parts of the problem set
- Solutions do not have to be perfect, but should include a discussion of all questions
- We encourage you to bring up any questions/issues you encountered during the problem set exercises
- Don't worry about the presentation format, e.g. your Jupyter notebook suffices to structure your discussion
Please comment extensively.
- Presentations are not graded, but we expect all students to attend and actively participate in the discussion

In-class presentations - Team assignment

Debug Sessions	Team Assignment
IO 1	DiDnt
June 12th	The Uninvertibles
	visible-hand
IO 2	invisible-hand
June 19th	Running LATE
	2SLS Crew
IO 3	GünTheLeo
June 26th	C's get degrees
	twosls
Labor 1	fuzzyRDD
July 3rd	BLUE
	mostly-harmless-rms
Labor 2	3-way-FE
July 10th	Praying Mantis
	MAL